

```
[52]: import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
```

```
[53]: df=pd.read_csv("titanic.csv") #load data set
```

```
[54]: df.head() #show first 5 rows
```

	PassengerId	Survived	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
[55]: df.shape #no of rows and columns
```

```
[55]: (891, 12)
```

```
[56]: df.duplicated().sum() #no of duplicate records
```

```
[56]: np.int64(0)
```

```
[57]: df = df.drop_duplicates() #remove duplicate records but this data not contain duplicate df
```

	PassengerId	Survived	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
--	-------------	----------	--------	------	--------	-----	-------	-------	--------	------	-------	----------

0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
---	---	---	---	-------------------------	------	------	---	---	-----------	--------	-----	---

0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Doolley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows x 12 columns

[58]: df.isnull().sum()

[58]:

PassengerId	0
Survived	0
Pclass	0
Name	0
Gender	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

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```
[59]: df[df['Age'].isnull()].index
```

what index is missing in Age

```
[59]: Index([ 5, 17, 19, 26, 28, 29, 31, 32, 36, 42,
```

```
...
      832, 837, 839, 846, 849, 859, 863, 868, 878, 888],
      dtype='int64', length=177)
```

```
[60]: df['Age']=df['Age'].fillna(df['Age'].mean()) #missing..numerical..mean
```

```
df['Age'].head(7)
```

#index[5] missing

```
[60]:
```

```
0    22.000000
1    38.000000
2    26.000000
3    35.000000
4    35.000000
5    29.699118
6    54.000000
```

Name: Age, dtype: float64

```
[61]: df[df['Embarked'].isnull()].index
```

#what index is missing in Embarked

```
[61]: Index([61, 829], dtype='int64')
```

```
[62]: df['Embarked']=df['Embarked'].fillna(df['Embarked'].mode()[0]) #missing..string..mode
```

```
df['Embarked'].head(63)
```

#index[61] missing

```
[62]:
```

```
0    S
1    C
2    S
3    S
4    S
```

```
58    S
59    S
60    C
61    S
```



[63]:

df[df['Cabin'].isnull()].index

#high missing data so we remove

[63]:

Index([0, 2, 4, 5, 7, 8, 9, 12, 13, 14, ...
878, 880, 881, 882, 883, 884, 885, 886, 888, 890],
dtype='int64', length=687)

[64]:

if 'Cabin' in df.columns:
df.drop('Cabin', axis=1, inplace=True)

[65]:

df

[65]:

	PassengerId	Survived	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.000000	1	0	A/5 21171	7.2500	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.000000	1	0	PC 17599	71.2833	C
2	3	1	3	Heikinen, Miss. Laina	female	26.000000	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.000000	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.000000	0	0	373450	8.0500	S
...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.000000	0	0	211536	13.0000	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.000000	0	0	112053	30.0000	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	29.699118	1	2	W./C. 6607	23.4500	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.000000	0	0	111369	30.0000	C
890	891	0	3	Dooley, Mr. Patrick	male	32.000000	0	0	370376	7.7500	Q

891 rows x 11 columns

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```
[66]: df.drop(['PassengerId', 'Name', 'Ticket'], axis=1, inplace=True)
df
```

[66]:

	Survived	Pclass	Gender	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22.000000	1	0	7.2500	S
1	1	1	female	38.000000	1	0	71.2833	C
2	1	3	female	26.000000	0	0	7.9250	S
3	1	1	female	35.000000	1	0	53.1000	S
4	0	3	male	35.000000	0	0	8.0500	S
...	...	...	...	...	...	...	...	...
886	0	2	male	27.000000	0	0	13.0000	S
887	1	1	female	19.000000	0	0	30.0000	S
888	0	3	female	29.699118	1	2	23.4500	S

```
[67]: (df['Age'] > 0) & (df['Age'] < 80)
```

[67]:

0	True
1	True
2	True
3	True
4	True
...	...
886	True
887	True
888	True
889	True
890	True

Name: Age, length: 891, dtype: bool

```
[68]: df = df[(df['Age'] > 0) & (df['Age'] < 80)]
df
```

#remove noise

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name: Age, length: 891, dtype: bool

```
[68]: df = df[(df['Age'] > 0) & (df['Age'] < 80)]  
df
```

	Survived	Pclass	Gender	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22.000000	1	0	7.2500	S
1	1	1	female	38.000000	1	0	71.2833	C
2	1	3	female	26.000000	0	0	7.9250	S
3	1	1	female	35.000000	1	0	53.1000	S
4	0	3	male	35.000000	0	0	8.0500	S
...	...	...	...	...	...	...	...	...
886	0	2	male	27.000000	0	0	13.0000	S
887	1	1	female	19.000000	0	0	30.0000	S
888	0	3	female	29.699118	1	2	23.4500	S
889	1	1	male	26.000000	0	0	30.0000	C
890	0	3	male	32.000000	0	0	7.7500	Q

890 rows x 8 columns

```
[69]: corr_matrix = df.corr(numeric_only=True)
```

#calculate correlation

```
[70]: to_drop = set()
```

empty set to store columns we delete

```
[71]: for i in range(len(corr_matrix.columns)):  
    for j in range(i):  
        if abs(corr_matrix.iloc[i, j]) >= 0.8:  
            colname = corr_matrix.columns[i]  
            to_drop.add(colname)
```

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[68]:

	Survived	Pclass	Gender	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22.000000	1	0	7.2500	S
1	1	1	female	38.000000	1	0	71.2833	C
2	1	3	female	26.000000	0	0	7.9250	S
3	1	1	female	35.000000	1	0	53.1000	S
4	0	3	male	35.000000	0	0	8.0500	S
...	...	...	...	...	...	...	...	...
886	0	2	male	27.000000	0	0	13.0000	S
887	1	1	female	19.000000	0	0	30.0000	S
888	0	3	female	29.699118	1	2	23.4500	S
889	1	1	male	26.000000	0	0	30.0000	C
890	0	3	male	32.000000	0	0	7.7500	Q

890 rows x 8 columns

```
[69]: corr_matrix = df.corr(numeric_only=True)
```

#calculate correlation

```
[70]: to_drop = set()
```

empty set to store columns we delete

```
[71]: for i in range(len(corr_matrix.columns)):
```

```
    for j in range(i):
        if abs(corr_matrix.iloc[i, j]) >= 0.8:
            colname = corr_matrix.columns[i]
            to_drop.add(colname)
```

```
df.drop(columns=to_drop, inplace=True)
```

```
df.corr(numeric_only=True)
```

st:8889/notebooks/Data%20mining%201.ipynb?

Toolbox

v4.20.0

conda

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Code

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Jupyterlab

[71]:

	Survived	Pclass	Age	SibSp	Parch	Fare
Survived	1.000000	-0.337020	-0.076030	-0.034683	0.082388	0.257602
Pclass	-0.337020	1.000000	-0.327746	0.082370	0.017636	-0.550337
Age	-0.076030	-0.327746	1.000000	-0.232556	-0.178663	0.092542
SibSp	-0.034683	0.082370	-0.232556	1.000000	0.414690	0.159648
Parch	0.082388	0.017636	-0.178663	0.414690	1.000000	0.216229
Fare	0.257602	-0.550337	0.092542	0.159648	0.216229	1.000000

[72]:

```
df['Age_group'] = pd.cut(df['Age'],
                           bins=[0, 12, 20, 40, 60, 80],
                           labels=['Child', 'Teen', 'Adult', 'Mid-age', 'Old'])
df.drop('Age', axis=1)
```

#discretization

anaconda

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- new notebook
- code snippets
- code snippets
- anaconda AI Assistant
- python help

[72]:

	Survived	Pclass	Gender	Age	SibSp	Parch	Fare	Embarked	Age_group
0	0	3	male	22.000000	1	0	7.2500	S	Adult
1	1	1	female	38.000000	1	0	71.2833	C	Adult
2	1	3	female	26.000000	0	0	7.9250	S	Adult
3	1	1	female	35.000000	1	0	53.1000	S	Adult
4	0	3	male	35.000000	0	0	8.0500	S	Adult
...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.000000	0	0	13.0000	S	Adult
887	1	1	female	19.000000	0	0	30.0000	S	Teen
888	0	3	female	29.699118	1	2	23.4500	S	Adult
889	1	1	male	26.000000	0	0	30.0000	C	Adult
890	0	3	male	32.000000	0	0	7.7500	Q	Adult

890 rows x 9 columns

```
[73]: df = pd.get_dummies(df, drop_first=True) #convert cat to string

[74]: train_df, test_df = train_test_split(df, test_size=0.3, random_state=42) #split dataset

[24]: train_df.to_csv("train_data.csv", index=False)
test_df.to_csv("test_data.csv", index=False)

print("Data preprocessing completed successfully!")

Data preprocessing completed successfully!
```